

Quiz-2 (Set A)

Name: _____ Roll: _____ Date: 03/11/2025

Instructions

1. You are given 60 minutes to answer 11 questions.
2. Please read all questions carefully before attempting.
3. There are 9 MCQ, each having four options, with only one correct answer. Select the best possible answer.
4. Each correct response earns 2 marks and incorrect answer earns -1.
5. There are two subjective questions, write your answer within the given space.

All The Best!

1. In the algorithm using min-hash over $[0,1]$, what is the expected minimum hash value for k distinct elements?

☐ $\frac{1}{k}$	☐ $\frac{1}{k} + 1$	☐ $\frac{1}{\sqrt{k}}$	☒ $\frac{1}{k+1}$
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2. For a stream of length n , in Misra-Gries with k counters, if an item survives till end then it implies that the,

☐ true frequency of the item is at least n/k	☐ true frequency of the item is at least of k^{th} most frequent item	☒ estimated frequency $\geq$ true frequency - n/k	☐ estimated frequency $\leq$ true frequency + n/k
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3. As dimension d increases such that it tends to ∞ , the volume of a unit sphere in R^d tends to

☐ Infinity	☒ Zero	☐ One	☐ Cannot Say
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4. When estimating distinct elements in the union of two streams using shared hash functions, which operation is used?

☐ Max of min-hash values	☐ Average of hash values	☒ Min of min-hash values	☐ Min of max-hash values
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5. Which of the following best explains why most volume of a high-dimensional sphere lies near its surface.

☐ Sphere becomes flat	☐ Surface area grows exponentially	☒ Inner volume shrinks faster than surface	☐ Radius increases with dimension
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6. In high dimensions, the expected angle between two random unit vectors tends to

☐ 0	☒ 90	☐ 180	☐ Cannot Say
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7. Which of the following is a limitation of Flajolet-Martin sketches?

☒ Cannot handle deletions	☐ Requires multiple passes over data	☐ Requires exact frequency counts	☐ Cannot be parallelized
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8. Recall faster matrix product using random vectors (not random sampling). Let A and B are two matrices of sizes 100×1000 and 1000×100 respectively such that $\text{rank}(A) = \text{rank}(B) = \text{rank}(AB) = 10$. Now, if V be a random vector in R^{1000} such that $E[VV^T] = I$, then what is the maximum possible rank C , where $C = AVV^TB$.

☐ 1000	☐ 100	☐ 10	☒ 1
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9. Consider a family of JL matrices $\mathcal{M}$ of size $p \times q$ such that for a set of vectors V in R^q , a random sample M from $\mathcal{M}$ ensures $(1 - \epsilon)\|v\|_2^2 \leq \|Mv\|_2^2 \leq (1 + \epsilon)\|v\|_2^2$ for every v in V with a high probability. Every index of M is iid sample from distribution of μ as mean and σ as the standard deviation. Which of the following is correct?

☒ $\mu = 0, \sigma = \frac{1}{\sqrt{p}}$	☐ $\mu = 0, \sigma = \frac{1}{q}$	☐ $\mu = 0, \sigma = \frac{1}{\sqrt{q}}$	☐ $\mu = 0, \sigma = \frac{1}{p}$
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10. Fill in the blanks. You may use the given box for calculating the asked value.

- a. Let X and Y be random vectors in d -dimensional space, such that each index of the vectors are independent random variables with uniform distribution in the set $\{-1, +1\}$. The expected squared Euclidian distance between them is, $E[\|X - Y\|_2^2] = \underline{2d}$. (7 Marks)

We know, $\|X - Y\|_2^2 = \|X\|_2^2 + \|Y\|_2^2 - 2X^T Y$.

Compute $E[\|X\|_2^2] = E[X^T X] = E[\sum_{i=1}^d X_i^2] = \sum_{i=1}^d E[X_i^2] = d$. Similarly, compute $E[\|Y\|_2^2] = d$

Compute $E[X^T Y] = E[\sum_{i=1}^d X_i Y_i] = \sum_{i=1}^d E[X_i Y_i] = \sum_{i=1}^d E[X_i] E[Y_i] = 0$.

Hence, $E[\|X - Y\|_2^2] = 2d$.

Rubric:

2 marks for identifying $\|X\|_2^2$, $\|Y\|_2^2$ and $X^T Y$ whose expectations has to be computed.

2 marks for computing expectation of $\|X\|_2^2$ or $\|Y\|_2^2$, since, $E[\|X\|_2^2] = E[\|Y\|_2^2]$

3 marks for computing expectation of $X^T Y$.

11. Recall, JL matrix of size $k \times n$ such that n -dimensional matrices are projected on to a k -dimensional space while preserving L_2 norms of the vectors. Define a random variable such that it gives a sparse JL where 80% of indices are zero. Show its correctness. (5 Marks)

Let the JL matrix be M , where, every index of M is sampled from the following distribution.

$$m = \begin{cases} -\sqrt{\frac{5}{k}} & \text{with probability } 0.1 \\ +\sqrt{\frac{5}{k}} & \text{with probability } 0.1 \\ 0 & \text{with probability } 0.8 \end{cases}$$

The students need to show that $E[M^T M] = I$

Rubric:

3 marks for correct definition of random variable. The random variable m is correct if $E[m] = 0$ and $\text{Var}(m) = 1/k$.

2 marks for showing that $E[M^T M] = I$